

Comparison & Logical Operators in SQL

Filtering, evaluating, and combining conditions in SQL queries with WHERE-clause examples and truth-table references.

MODULE

Querying & Filtering Data

FORMAT

SQL Theory + Practice

What this report covers

01

Comparison Operators

=, <>, >, <, >=, <= - comparing values in WHERE clauses

02

Logical Operators

AND, OR, NOT - combining multiple conditions

03

Worked Examples

Practical queries on a sample Employees table

04

Truth Tables & Quick Reference

Summary tables for revision before exams

Why Operators Matter in SQL

SQL operators let you compare values and combine multiple conditions inside a WHERE clause, so a query returns exactly the rows that meet your criteria. They fall into two main families covered in this report: comparison operators, which test a relationship between two values, and logical operators, which combine the results of multiple conditions.

Comparison Operators

Used to compare a column's value against another value or column. Each comparison evaluates to TRUE, FALSE, or UNKNOWN.

= <> > < >= <=

Logical Operators

Used to combine two or more conditions, or to reverse a condition's result, inside a single WHERE clause.

AND OR NOT

Sample Table Used in Examples: Employees

emp_id	name	department	salary	age
101	Aarav Shah	Sales	45000	29
102	Priya Nair	IT	62000	34
103	Karan Mehta	IT	58000	41
104	Sneha Rao	HR	39000	26
105	Vikram Joshi	Sales	51000	37

Comparison Operators - Reference

Each operator below tests how the value on its left relates to the value on its right. The result is always TRUE, FALSE, or UNKNOWN (when NULL is involved).

OPERATOR	MEANING	EXAMPLE & EXPLANATION
=	Equal to	<code>salary = 45000</code> Matches rows where salary is exactly 45000
<> or !=	Not equal to	<code>department <> 'HR'</code> Excludes all HR department rows
>	Greater than	<code>salary > 50000</code> Employees earning more than 50,000
<	Less than	<code>age < 30</code> Employees younger than 30 years
>=	Greater than or equal to	<code>salary >= 45000</code> Includes 45,000 and above
<=	Less than or equal to	<code>age <= 30</code> Includes age 30 and below

Comparison Operators - Worked Examples

EXAMPLE 1

Find employees earning more than 50,000

QUERY.SQL

```
SELECT name, salary
FROM Employees
WHERE salary > 50000;
```

```
-- Returns: Priya Nair, Vikram Joshi
```

EXAMPLE 2

Find employees who are NOT in the IT department

QUERY.SQL

```
SELECT name, department
FROM Employees
WHERE department <> 'IT';
```

```
-- Returns: Aarav Shah, Sneha Rao, Vikram Joshi
```

EXAMPLE 3

Find employees aged 30 or below

QUERY.SQL

```
SELECT name, age
FROM Employees
WHERE age <= 30;
```

```
-- Returns: Aarav Shah, Sneha Rao
```

TIP

Use single quotes for text values ('IT', 'HR') and no quotes for numbers (50000, 30).

Logical Operators - AND, OR, NOT

Logical operators combine the results of two or more comparison conditions into a single TRUE/FALSE outcome for each row, deciding whether that row appears in the final result.

AND

All conditions must be TRUE

```
department = 'IT' AND salary > 60000
```

OR

At least one condition must be TRUE

```
department = 'HR' OR department = 'Sales'
```

NOT

Reverses the result of a condition

```
NOT department = 'IT'
```

Truth Table - AND / OR

CONDITION A	CONDITION B	A AND B	A OR B
TRUE	TRUE	TRUE	TRUE
TRUE	FALSE	FALSE	TRUE
FALSE	TRUE	FALSE	TRUE
FALSE	FALSE	FALSE	FALSE

Logical Operators - Worked Examples

EXAMPLE 1

AND - IT employees earning above 60,000

QUERY.SQL

```
SELECT name, department, salary
FROM Employees
WHERE department = 'IT'
      AND salary > 60000;

-- Returns: Priya Nair
```

EXAMPLE 2

OR - employees in HR or Sales department

QUERY.SQL

```
SELECT name, department
FROM Employees
WHERE department = 'HR'
      OR department = 'Sales';

-- Returns: Aarav Shah, Sneha Rao, Vikram Joshi
```

EXAMPLE 3

NOT - employees who are not in IT

QUERY.SQL

```
SELECT name, department
FROM Employees
WHERE NOT department = 'IT';

-- Returns: Aarav Shah, Sneha Rao, Vikram Joshi
```

COMBINING OPERATORS

```
WHERE (department = 'IT' OR department = 'Sales')
      AND salary > 45000;
```

Quick Reference & Key Takeaways

Comparison Operators

- = checks exact equality between two values
- <> or != checks that two values are different
- > and < check strictly greater / strictly less than
- >= and <= include the boundary value itself

Logical Operators

- AND narrows results - every condition must be TRUE
- OR widens results - only one condition needs to be TRUE
- NOT reverses a condition's result, turning TRUE into FALSE and vice versa
- Parentheses control the order conditions are evaluated, just like in mathematics

One-Line Cheat Sheet

```
WHERE col1 >= value1 AND (col2 = 'X' OR col2 = 'Y')
```

Why this matters

Comparison and logical operators are the foundation of the WHERE clause. Mastering them is essential before moving on to more advanced filtering with BETWEEN, IN, LIKE, and subqueries.